MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 10

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Continuous-Time Dynamics

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\boldsymbol{\psi}_t(\mathbf{x}_0) = \int_{t_0}^{t} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(s), s) ds + \mathbf{x}_0 \approx \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0, t).$$

- $\mathbf{v}_{\boldsymbol{\theta}} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a vector field.
- $\boldsymbol{\psi} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a flow (the solution of ODE):

Euler `ODESolve`

$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

For a smooth field on a fixed interval, compare global error and **NFE** (velocity-network evaluations):

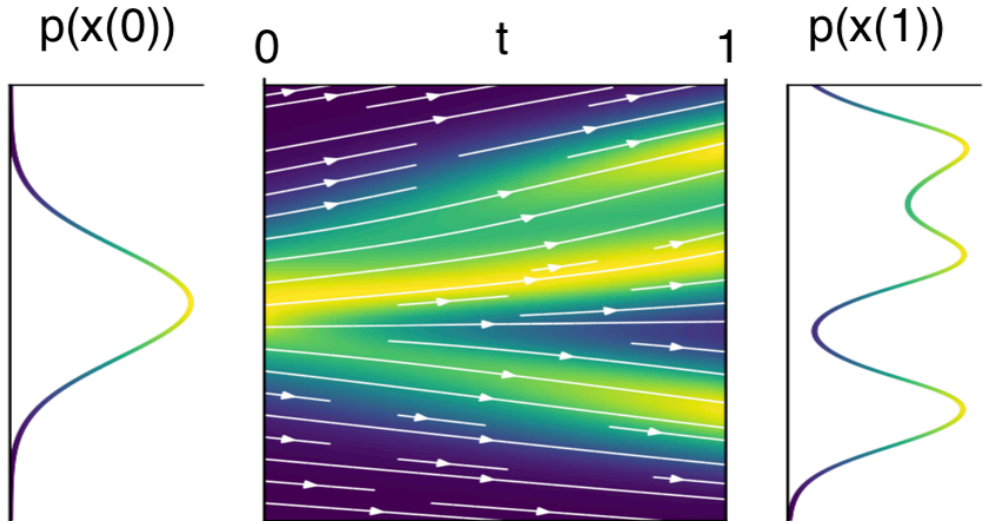
- **Euler:** $\mathcal{O}(h)$ global error, **1** evaluation per step.
- **Heun:** $\mathcal{O}(h^2)$ global error, **2** evaluations per step, averaging the current and predicted endpoint velocities.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Driscoll T. A., Braun R. J. Fundamentals of Numerical Computation: Runge–Kutta methods

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- Suppose $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$. Then, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.
- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.



Recap of Previous Lecture

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

This guarantees the ODE is **uniquely reversible**.

$$\boldsymbol{\psi}_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t)dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)dt$$

Note: Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection).

How can we compute $p_t(\mathbf{x})$ at arbitrary t?

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t, and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

Hutchinson's trace estimator:

$$\operatorname{tr}(\mathbf{J}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})}[\boldsymbol{\epsilon}^T \mathbf{J} \boldsymbol{\epsilon}], \qquad \mathbf{J} = \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}.$$

Exact density identity; approximate computation. Use a numerical ODE solver and one noise vector for an unbiased trace estimate via autodiff.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

Recap of Previous Lecture

Fix $p_0 = \mathcal{N}(0, \mathbf{I})$ and learn $\mathbf{v}_{\boldsymbol{\theta}}$; the endpoint density is $p_1 = p_{\boldsymbol{\theta}}$.

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}$. Solve the ODE **backward**, from $\mathbf{x}(1) = \mathbf{x}$ to $\mathbf{x}(0)$.
2. Compute the negative log-likelihood along this trajectory:

$$\mathcal{L} = -\log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\log p_0(\mathbf{x}(0)) + \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

3. Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}$; differentiate through the ODE, e.g., via the **adjoint method**.

Sampling

1. Sample $\mathbf{x}(0) \sim p_0$.
2. Solve the ODE **forward** and return the endpoint:

$$\mathbf{x}(1) = \texttt{ODESolve}_v(\mathbf{x}(0), \boldsymbol{\theta}, t_0 = 0, t_1 = 1) \sim p_{\boldsymbol{\theta}}.$$

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

Recap of Previous Lecture

SDE Basics

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where $\mathbf{w}(t)$ is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I}), \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \text{ where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Discretizing the SDE (Euler-Maruyama Update) – `SDESolve`

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \boldsymbol{\epsilon} \cdot \sqrt{dt}$$

- At each time t, we have the density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \to \mathbb{R}_+$ is a **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.

Recap of Previous Lecture

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}} p_t(\mathbf{x})$$

Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2}\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})dt + 1 \cdot d\mathbf{w}$$

The density $p_t(\mathbf{x})$ is a **stationary** distribution for the SDE. Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$ for all t:

Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Outline

01 **Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)**

02 Probability Flow ODE

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Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$
$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1}, \quad q(\mathbf{x}_{t-1}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$
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$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Let's transform this Markov chain into the continuous stochastic process $\mathbf{x}(t)$ by letting $T \to \infty$:

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon}$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$
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$$\begin{aligned}\mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt} dt} \cdot \boldsymbol{\epsilon}\end{aligned}$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$
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Let's transform this Markov chain into the continuous stochastic process $\mathbf{x}(t)$ by letting $T \to \infty$:

$$\begin{aligned}
\mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt} dt} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}
\end{aligned}$$

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$ is a monotonically increasing function.

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$ is a monotonically increasing function.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting $T \to \infty$ and setting $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$ (where $dt = \frac{1}{T}$):

$$\mathbf{x}(t) = \sqrt{1-\beta(t)dt} \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting $T \to \infty$ and setting $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$ (where $dt = \frac{1}{T}$):

$$\begin{aligned} \mathbf{x}(t) &= \sqrt{1-\beta(t)dt} \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \\ &\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \end{aligned}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

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$$\begin{aligned}
\mathbf{x}(t) &= \sqrt{1-\beta(t)dt} \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \\
&\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t-dt) - \frac{1}{2}\beta(t)\mathbf{x}(t-dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}
\end{aligned}$$

Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

We treat the discrete-time methods (NCSN, DDPM) as a special case of the continuous-time methods (VE-SDE, VP-SDE).

Outline

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Probability Flow ODE

ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution $p_0(\mathbf{x})$.

Probability Flow ODE

ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution $p_0(\mathbf{x})$.

SDE and KFP Equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}} p_t(\mathbf{x})$$

Now there are two sources of randomness: the initial distribution $p_0(\mathbf{x})$ and the Wiener process $\mathbf{w}(t)$.

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

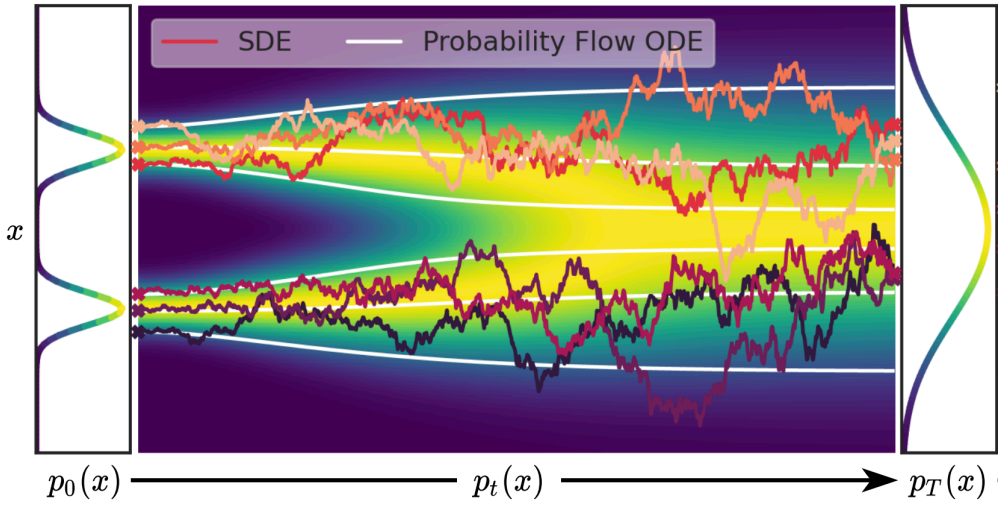
$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

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Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

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Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right)$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

Proof

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) \\ &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}\right]\right)\end{aligned}$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

Proof

$$\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right]\right)
\end{aligned}$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

Proof

$$\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right)
\end{aligned}$$

Probability Flow ODE

Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) p_t(\mathbf{x})\right]\right)$$

Probability Flow ODE

Proof (Continued)

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right) \\ &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})\right]\right) = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})\right)\end{aligned}$$

Probability Flow ODE

Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right)$$

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$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x}); \quad \tilde{g}(t) = 0$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt + 0 \cdot d\mathbf{w} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}\tilde{g}^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Probability Flow ODE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

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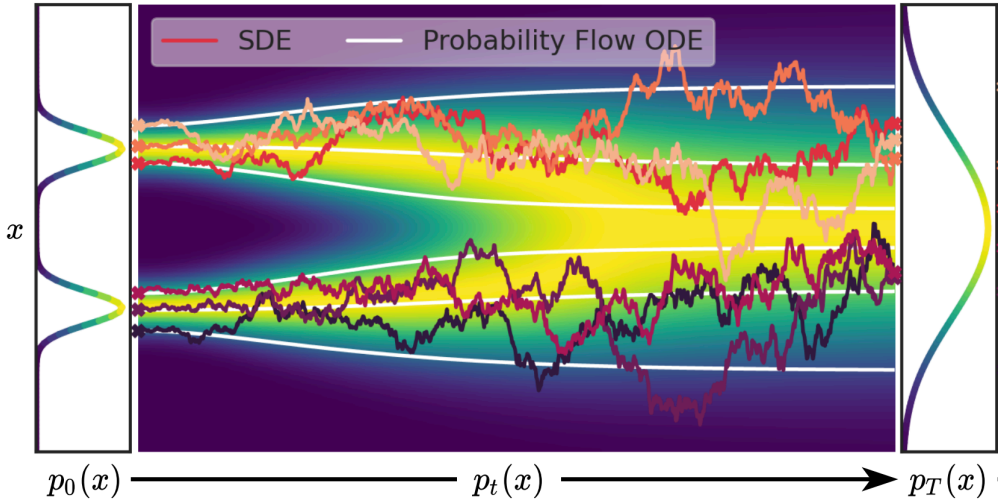
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- $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ is the continuous-time score function.
- The ODE produces more stable trajectories.



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Why convert SDEs to PF-ODEs? (1) Efficient Sampling

Use the familiar ODE solvers with the learned PF-ODE field:

$$\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t).$$

Solver	How it uses the network
Euler / Heun	Apply the familiar ODE updates: 1 / 2 evaluations per step. EDM uses Heun.

Karras T., et al. Elucidating the Design Space of Diffusion-Based Generative Models, 2022
Lu C., et al. DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps, 2022
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Compare **NFE** (network evaluations). DPM-Solver reports **10–20 NFE**; the required budget depends on the model.

Noise injection is a separate choice: it can improve quality, but does not help every model.

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Why convert SDEs to PF-ODEs? (2) Deterministic Sampling

- The PF-ODE defines a deterministic, invertible map $\mathbf{x}(0) \leftrightarrow \mathbf{x}(T)$.
- Each terminal $\mathbf{x}(T)$ produces a unique trajectory back to data space.

Chen R. T. Q., et al. Neural Ordinary Differential Equations, 2018
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Log-likelihood via the probability-flow ODE

$$\log p_0(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt$$

Exact change-of-variables identity; numerical evaluation uses an ODE solver and, optionally, Hutchinson's trace estimator.

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- Connects PF-ODE to continuous normalizing flows.
- Invertibility enables image $\rightarrow$ latent $\rightarrow$ edited image workflows.
- Smooth interpolation in latent space between two samples.

Chen R. T. Q., et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

Sampling: Reverse SDE vs PF-ODE

The same trained score $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)$ defines two sampling dynamics (reverse SDE derived next).

Sampling: Reverse SDE vs PF-ODE

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Sampling (Reverse SDE, Euler–Maruyama)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Initialize step $\Delta t = T/K$.
3. Denoise:

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Big[\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\,\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\Big]\Delta t + g(t)\sqrt{\Delta t}\cdot\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$$

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Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Reverse SDE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \qquad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Here dt can be > 0 or < 0.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
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Let $\tau = 1 - t$ ($d\tau = -dt$).

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Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

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- Here the score function appears: $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.
- Since $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$ (PF-ODE), the reverse drift equals $\mathbf{f} - g^2\mathbf{s} = \mathbf{v} - \frac{1}{2}g^2\mathbf{s}$.

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Proof Sketch

- Convert the initial SDE to a probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reversed probability flow ODE back to an SDE.

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Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

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$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1-\tau) + g^2(1-\tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1-\tau) d\mathbf{w}$$

Here $\mathbf{w}(\tau)$ is a Wiener process in forward τ-time.

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There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

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Proof (Continued)

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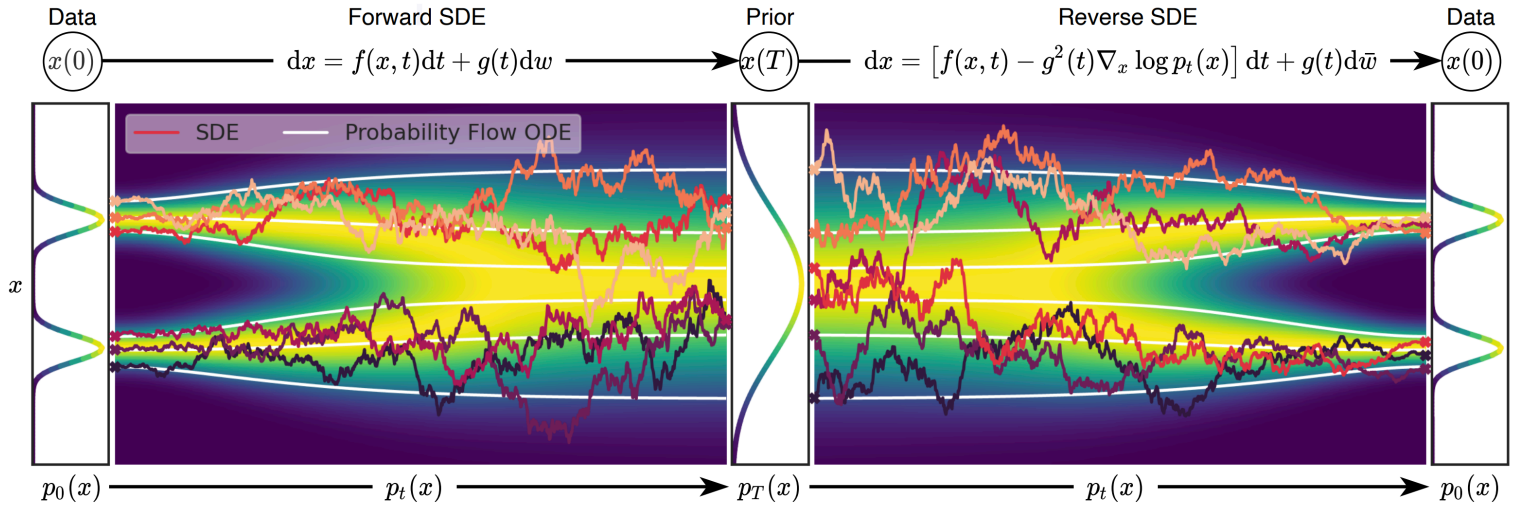
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VE and VP: Probability Flow and Reverse SDE

Write $\mathbf{s} = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

VE / NCSN

$\sigma(t)$ is monotonically increasing.

$$d\mathbf{x} = \sqrt{\frac{d\sigma^2(t)}{dt}}\, d\mathbf{w}$$

VP / DDPM

Unit initial variance is preserved.

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\, dt + \sqrt{\beta(t)}\, d\mathbf{w}$$

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Unit initial variance is preserved.

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$$d\mathbf{x} = \left(-\frac{1}{2}\beta(t)\mathbf{x} - \frac{1}{2}\beta(t)\mathbf{s}\right) dt$$

VE and VP: Probability Flow and Reverse SDE

Write $\mathbf{s} = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

VE / NCSN

$\sigma(t)$ is monotonically increasing.

$$d\mathbf{x} = \sqrt{\frac{d\sigma^2(t)}{dt}}\, d\mathbf{w}$$

Probability Flow ODE

$$d\mathbf{x} = -\frac{1}{2}\frac{d\sigma^2(t)}{dt}\mathbf{s}\, dt$$

Reverse SDE

$$d\mathbf{x} = -\frac{d\sigma^2(t)}{dt}\mathbf{s}\, dt + \sqrt{\frac{d\sigma^2(t)}{dt}}\, d\bar{\mathbf{w}}$$

Here $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

VP / DDPM

Unit initial variance is preserved.

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\, dt + \sqrt{\beta(t)}\, d\mathbf{w}$$

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Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Score-Based Generative Models Through SDEs

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

Score-Based Generative Models Through SDEs

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Score-Based Generative Models Through SDEs

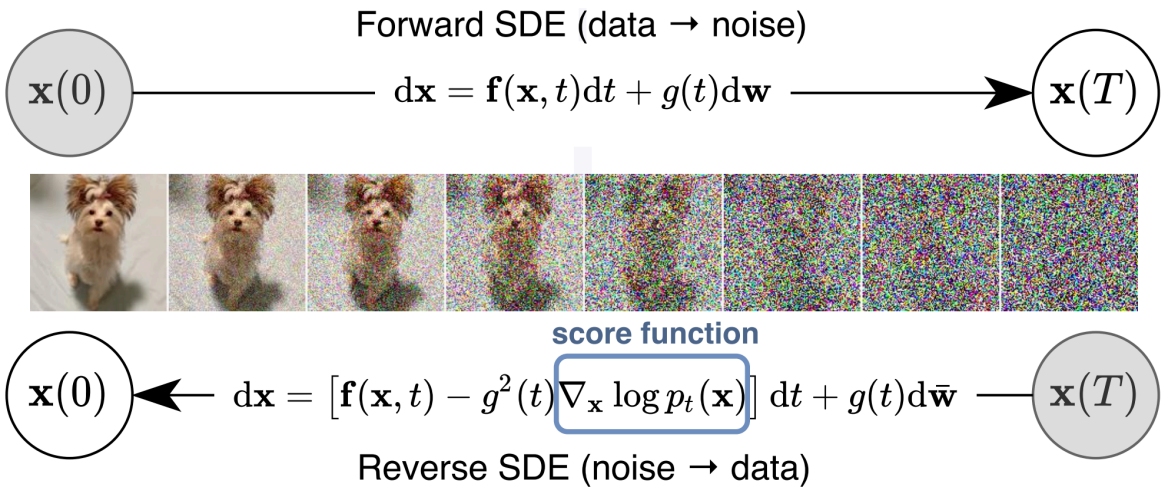
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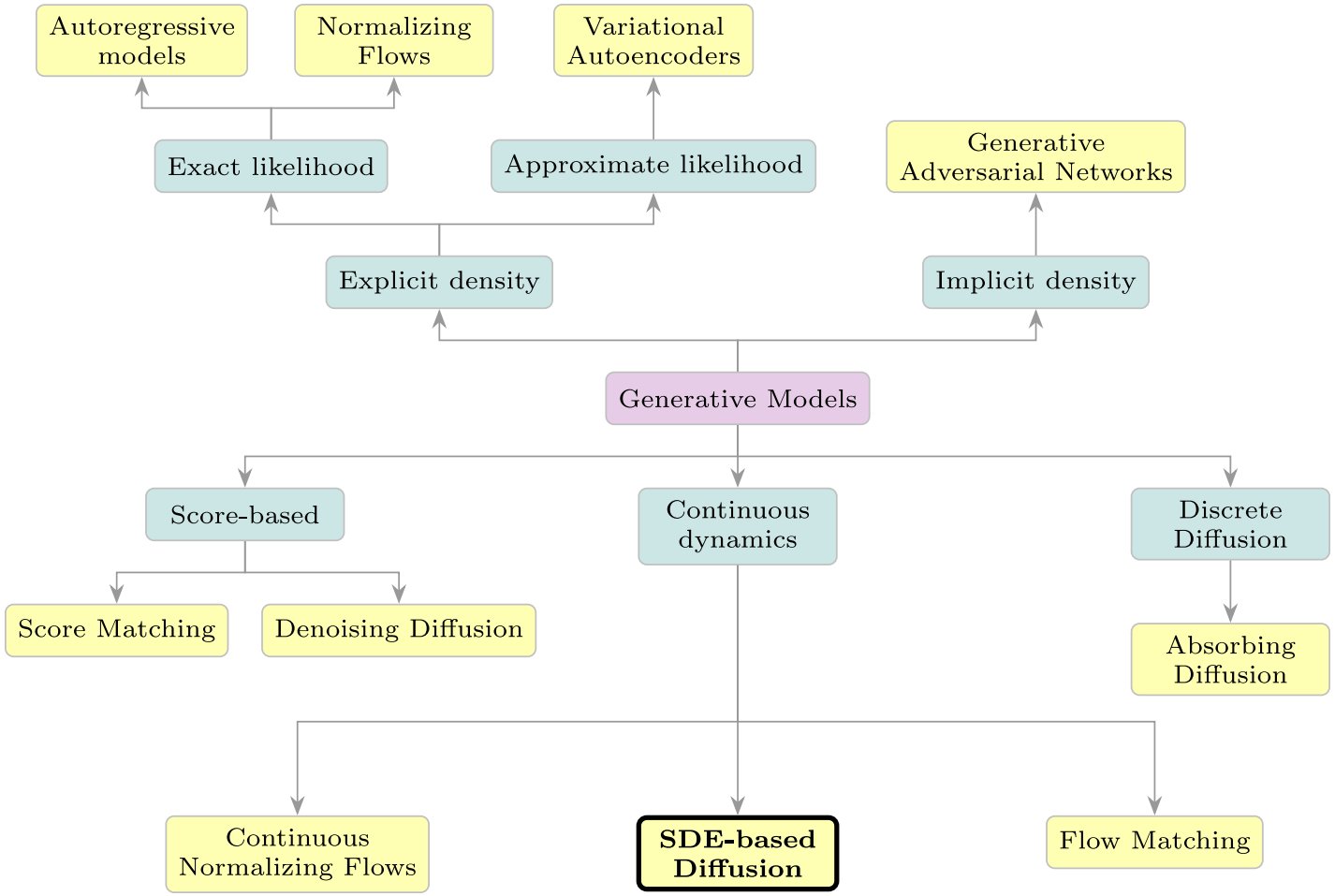
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Generative Models Taxonomy



Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t)-\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

Score-Based Generative Models Through SDEs

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$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0),t), \sigma^2(t)\cdot\mathbf{I}\Big)$$

$$\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\sigma^2(t)}(\mathbf{x}(t)-\boldsymbol{\mu})$$

Note: Normality holds for $\mathbf{f}(\mathbf{x},t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x},t) = f(t)\mathbf{x}$).

Score-Based Generative Models Through SDEs

Continuous-Time Objective

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$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}}\cdot d\mathbf{w} \;\; \text{(Variance Exploding SDE)}$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)}\cdot d\mathbf{w} \;\; \text{(Variance Preserving SDE)}$$

Is it possible to explicitly derive $\boldsymbol{\mu}(\mathbf{x}(0),t)$ and $\sigma^2(t)$ for VE-SDE and VP-SDE?

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\Big)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t), \qquad \frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t)$$

Score-Based Generative Models Through SDEs

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Proof

$$\mathbb{E}\left[d\mathbf{x}|\mathbf{x}(0)\right] = \mathbb{E}\left[f(t)\mathbf{x}dt|\mathbf{x}(0)\right] + \mathbb{E}\left[g(t)d\mathbf{w}|\mathbf{x}(0)\right]$$

We prove the mean equation only; the variance equation is stated without proof.

Score-Based Generative Models Through SDEs

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Score-Based Generative Models Through SDEs

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Score-Based Generative Models Through SDEs

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&= f(t)\mathbb{E}\left[\mathbf{x}|\mathbf{x}(0)\right]dt + g(t)\mathbb{E}\left[d\mathbf{w}|\mathbf{x}(0)\right] \\
&= f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)dt \\
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Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

Examples

$$\textbf{NCSN:} \quad f(t) = 0 \quad \Rightarrow \quad \boldsymbol{\mu} = \mathbf{x}(0)$$

Score-Based Generative Models Through SDEs

Theorem

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Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

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$$\boldsymbol{\mu} = \mathbf{x}(0)\exp\left(-\frac{1}{2}\int_0^t \beta(s)ds\right)$$

Score-Based Generative Models Through SDEs

Training

1. Sample $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$, $t \sim U[0, 1]$.
2. Sample $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$.
3. Compute loss $\mathcal{L} = \left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$.

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\Big)$$

Score-Based Generative Models Through SDEs

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NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right] \cdot \mathbf{I}\right)$$

Score-Based Generative Models Through SDEs

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DDPM

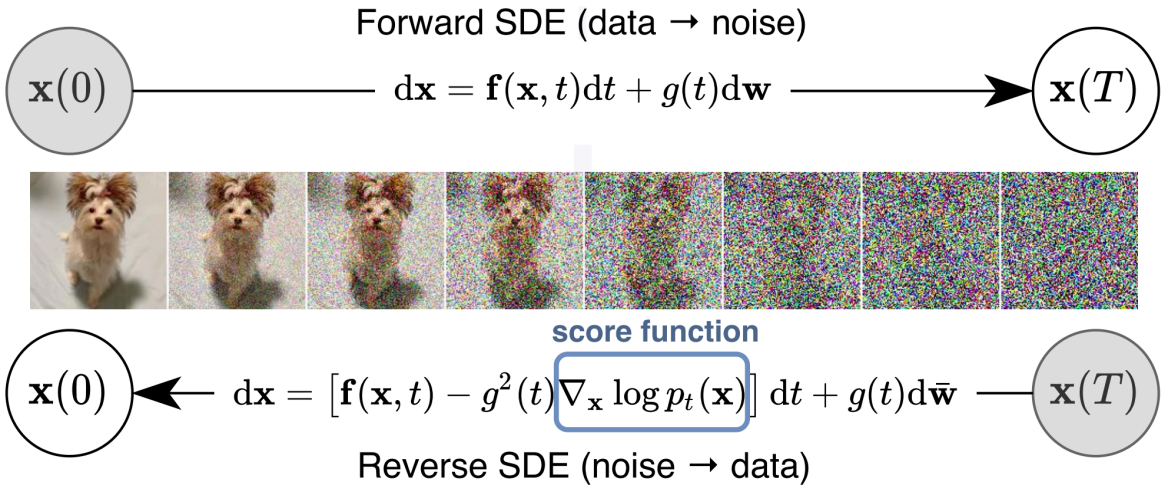
$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2}\int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Sampling with the Reverse SDE

Use the trained score for sampling. Assume terminal prior $\mathcal{N}(0, \mathbf{I})$.

Sampling

1. Sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the reverse SDE using numerical solvers (`SDESolve`).

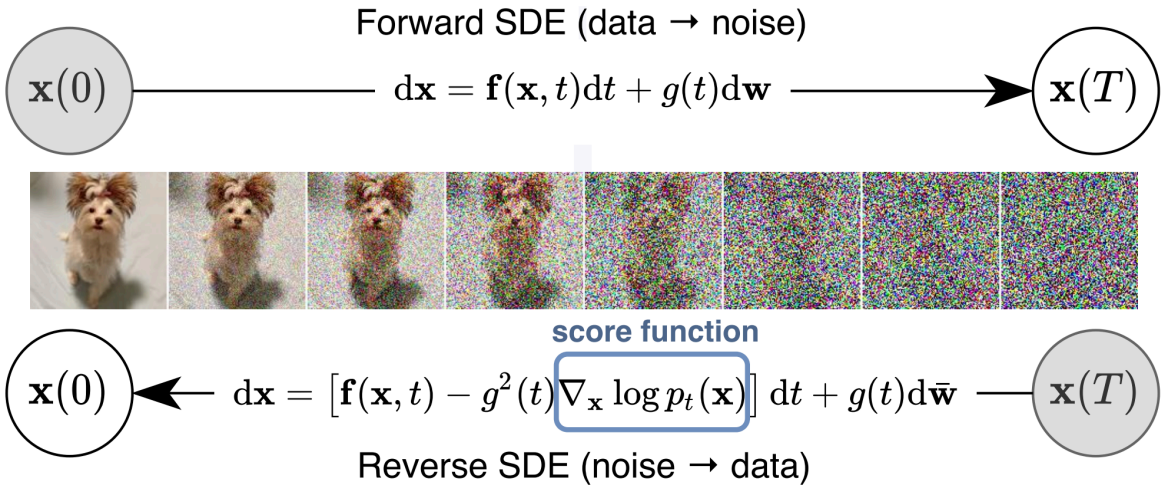


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- Discretizing the reverse SDE provides ancestral sampling.
- Discretizing the probability flow ODE yields deterministic sampling.

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

From Time to Noise Level

Use the familiar additive-noise form: $\mathbf{x} = \mathbf{x}_0 + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

$$\mathbf{f}(\mathbf{x}, t) = 0, \qquad g^2(t) = \frac{d\sigma^2(t)}{dt} = 2\sigma(t)\frac{d\sigma(t)}{dt}.$$

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Rewrite the probability flow ODE using σ

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{f}(\mathbf{x}(t), t) - \frac{1}{2}g^2(t)\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)) \\ &= -\sigma(t)\frac{d\sigma(t)}{dt}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)).\end{aligned}$$

$$\frac{d\mathbf{x}}{d\sigma} = \left(\frac{d\sigma(t)}{dt}\right)^{-1}\frac{d\mathbf{x}(t)}{dt} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma).$$

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$$\frac{d\mathbf{x}}{d\sigma} = \left(\frac{d\sigma(t)}{dt}\right)^{-1}\frac{d\mathbf{x}(t)}{dt} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma).$$

Choose a denoiser using Tweedie's formula

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}] = \mathbf{x} + \sigma^2\nabla_{\mathbf{x}}\log p_\sigma(\mathbf{x}).$$

$$D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) \approx \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}], \qquad \frac{d\mathbf{x}}{d\sigma} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) = \frac{\mathbf{x} - D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)}{\sigma}.$$

Decreasing σ moves towards the denoiser prediction. **How do we train this denoiser?**

Train the Denoiser

Use the existing score-training objective. For a noisy-clean pair $(\mathbf{x}, \mathbf{x}_0)$ with $q(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma^2 \mathbf{I})$:

$$\mathcal{L} = \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) \right\|_2^2 .$$

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Compute the conditional Gaussian target

$$\nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) = -\frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} .$$

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Compute the conditional Gaussian target

$$\nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) = -\frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} .$$

Substitute the denoiser representation

$$\begin{aligned} \mathcal{L} &= \left\| \frac{D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}}{\sigma^2} + \frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} \right\|_2^2 \\ &= \frac{1}{\sigma^4} \left\| D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}_0 \right\|_2^2 . \end{aligned}$$

Same training objective, different prediction target.
The trained denoiser now supplies the field for sampling.

Generate Samples with a Denoiser

Use the trained denoiser in the ODE solver. Start from strong noise and decrease σ.

Generate Samples with a Denoiser

Use the trained denoiser in the ODE solver. Start from strong noise and decrease σ.

Sampling (Euler)

1. Choose $\sigma_0 > \sigma_1 > \cdots > \sigma_N = 0$ and sample $\mathbf{x}^{(0)} \sim \mathcal{N}(0, \sigma_0^2 \mathbf{I})$.
2. For $i = 0, \ldots, N-1$, evaluate $D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)$ and update:

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + (\sigma_{i+1} - \sigma_i) \frac{\mathbf{x}^{(i)} - D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)}{\sigma_i}.$$

3. Return $\mathbf{x}^{(N)}$.

We now have an explicit Euler sampler in noise coordinates.
How does this relate to the deterministic DDIM update?

DDIM Is Euler in Noise Coordinates

Recall deterministic DDIM $(s < t)$:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \qquad \hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}},$$

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Recall the DDPM / NCSN rescaling ($\sigma_s < \sigma_t$)

$$\tilde{\mathbf{x}}_u = \frac{\mathbf{x}_u}{\sqrt{\bar{\alpha}_u}}, \qquad \sigma_u = \sqrt{\frac{1-\bar{\alpha}_u}{\bar{\alpha}_u}}, \qquad u \in \{s, t\}.$$

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$$D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t) = \hat{\mathbf{x}}_0 = \tilde{\mathbf{x}}_t - \sigma_t \hat{\boldsymbol{\epsilon}}.$$

Rewrite DDIM as the denoiser Euler step

$$\begin{aligned}\tilde{\mathbf{x}}_s &= \hat{\mathbf{x}}_0 + \sigma_s \hat{\boldsymbol{\epsilon}} = \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\hat{\boldsymbol{\epsilon}} \\ &= \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\frac{\tilde{\mathbf{x}}_t - D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t)}{\sigma_t}.\end{aligned}$$

DDIM implements this Euler step using the existing noise predictor.

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Continuous-Time Normalizing Flows

Now learn the velocity field itself. Recall $\mathbf{x}_t = \mathbf{x}(t)$ for $t \in [0, 1]$:

- $\mathbf{x}_0 \sim p_0(\mathbf{x})$, where $p_0(\mathbf{x})$ is a base distribution (mostly $\mathcal{N}(0, \mathbf{I})$);
- $\mathbf{x}_1 \sim p_1(\mathbf{x})$, where $p_1(\mathbf{x})$ is the true data distribution $p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity $\mathbf{v}$ determines a flow $\boldsymbol{\psi}_t$: $\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

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KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

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- Likelihood-based CNF training requires ODE solves and Jacobian trace evaluations.
- The adjoint method computes gradients through the ODE solution; repeated solves can be costly.

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- Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- Flow matching fits velocities by regression, without model ODE solves or trace evaluations during training.

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Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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- Approximate the true vector field $\mathbf{v}(\mathbf{x}, t)$ using $\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)$.
- Use the learned flow $\boldsymbol{\psi}_t$ for deterministic sampling: $\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0)$.

Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

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Training

1. Sample $t\sim U[0,1]$, $\mathbf{x}_t\sim p_t(\mathbf{x})$.
2. Compute loss $\mathcal{L}=\left\|\mathbf{v}(\mathbf{x}_t,t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

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Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Flow Matching: Training and Sampling

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Problem: The true vector field $\mathbf{v}(\mathbf{x},t)$ is **unknown**.

Flow Matching

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- There are infinitely many possible $\mathbf{v}(\mathbf{x},t)$ between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- We need to select the "best" $\mathbf{v}(\mathbf{x},t)$ and make the objective tractable.

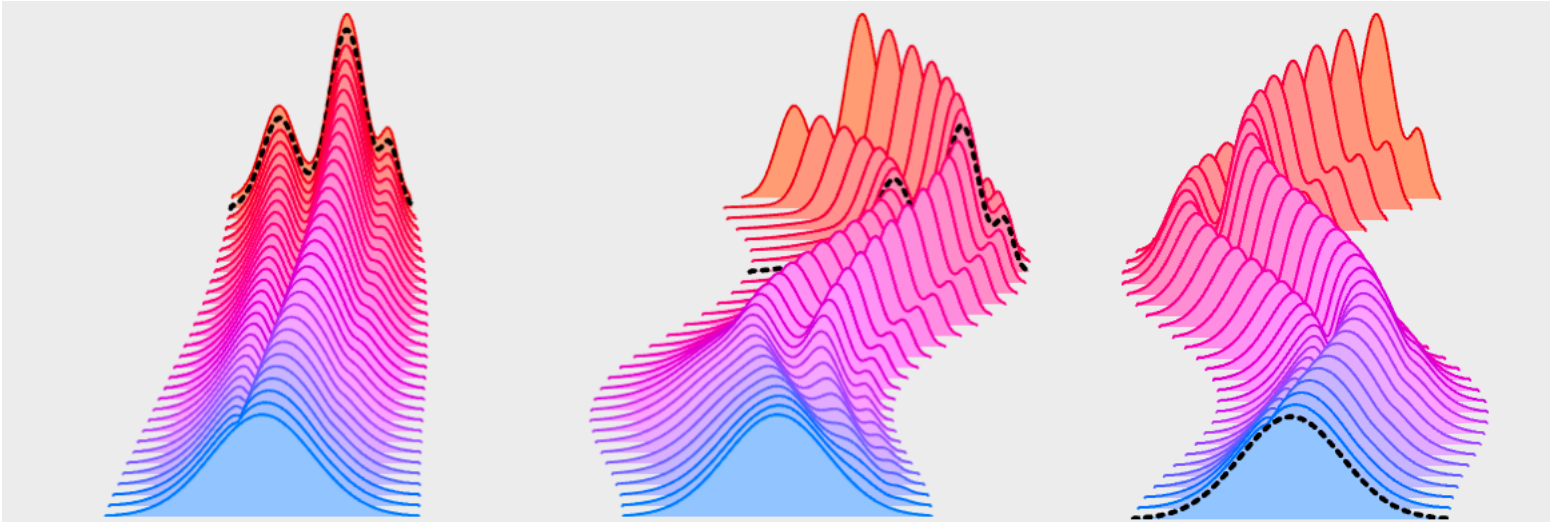


image credit: A Visual Dive into Conditional Flow Matching

Summary

- Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- Every SDE admits a corresponding probability flow ODE that follows the same probability path $p_t(\mathbf{x})$, yielding deterministic invertible sampling and log-likelihood evaluation via the instantaneous change of variables.
- A denoiser parameterizes the PF-ODE; DDIM is Euler in rescaled state and noise coordinates. Compare solver cost in network evaluations (NFE).
- SDEs can be reversed in time using the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$; discretizing the reverse SDE gives ancestral sampling, while discretizing the PF-ODE gives deterministic sampling.
- The continuous-time score matching objective generalizes DDPM and NCSN; for affine-drift SDEs the transition kernel $q(\mathbf{x}(t)|\mathbf{x}(0))$ is Gaussian, making the training objective tractable.
- Flow matching trains the velocity field by regression, avoiding model ODE solves and trace evaluations during training.